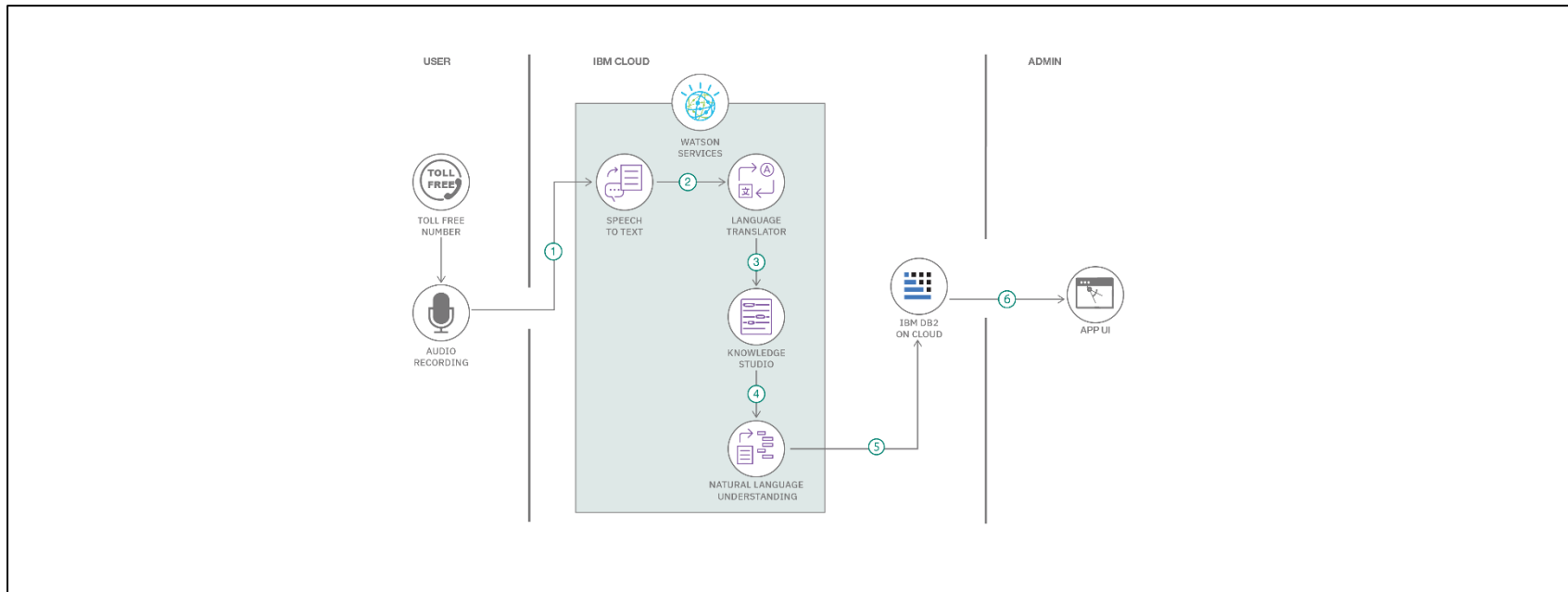


## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                                                                                     |
|---------------|-------------------------------------------------------------------------------------|
| Date          | 01 October 2026                                                                     |
| Team ID       | SWTID-2026-9203                                                                     |
| Project Name  | Import Data using Transform Maps<br>(Spreadsheet) – ServiceNow Employee Data Import |
| Maximum Marks | 4 Marks                                                                             |

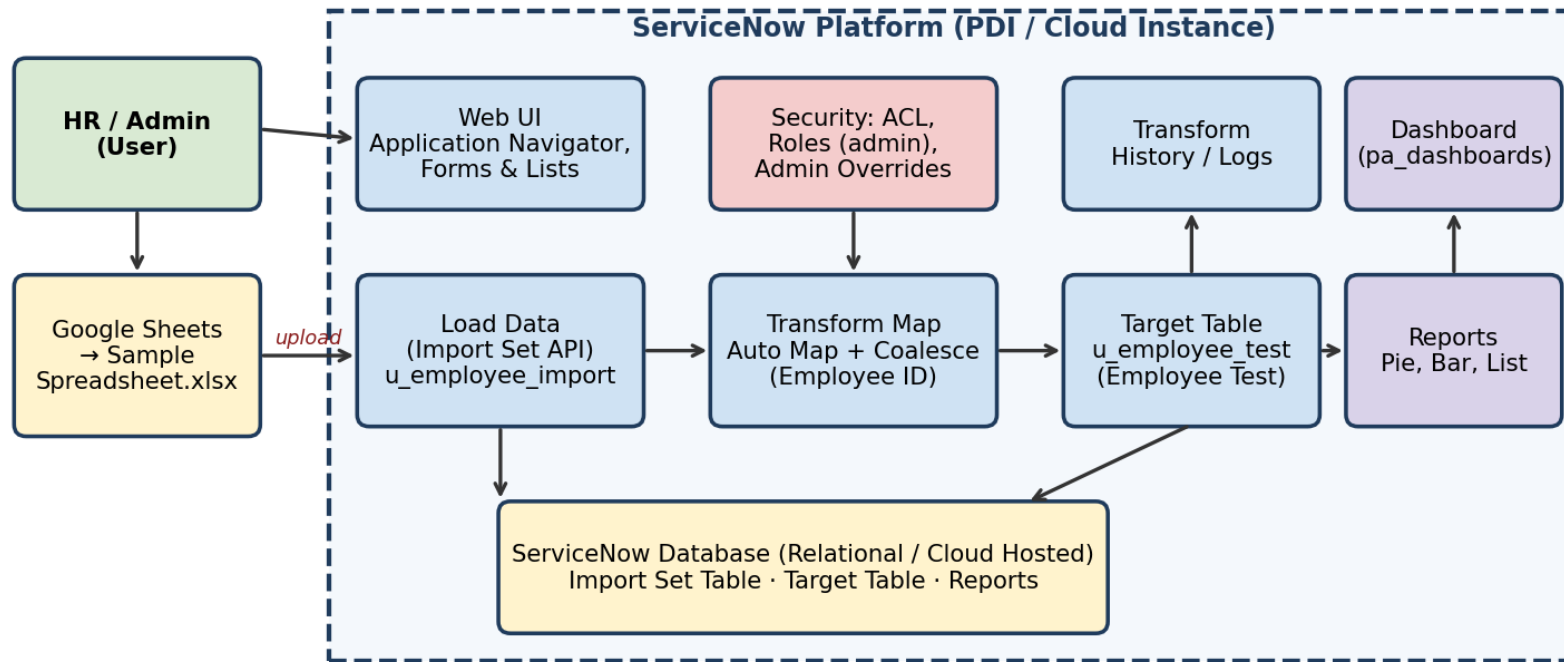
### Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2



**Our Project Architecture: Employee Data Import & Analytics on ServiceNow**

## Technical Architecture: Employee Data Import & Analytics on ServiceNow



**Table-1 : Components & Technologies:**

| <b>S.No</b> | <b>Component</b>       | <b>Description</b>                                                                                                                                              | <b>Technology</b>                                                |
|-------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1.          | User Interface         | How the admin / HR user interacts with the application: ServiceNow web UI (Application Navigator, Load Data page, forms, lists, reports and dashboards)         | ServiceNow Platform UI (HTML5, CSS, JavaScript) in a web browser |
| 2.          | Application Logic-1    | Import Set processing: the spreadsheet (Sample Spreadsheet.xlsx) is loaded into the staging table Employee Import (u_employee_import)                           | ServiceNow Load Data / Import Set                                |
| 3.          | Application Logic-2    | Transform Map "Sample Spreadsheet Import" maps source fields to Employee Test fields and uses Coalesce on Employee ID to insert new and update existing records | ServiceNow Transform Map (Auto Map, Mapping Assist, Coalesce)    |
| 4.          | Application Logic-3    | Reporting and dashboards: Employees by Department (Pie), Employees by Location (Bar), Employee List Report, shown on Employee Analytics Dashboards              | ServiceNow Reports & Dashboards (pa_dashboards)                  |
| 5.          | Database               | Import staging table u_employee_import and target table u_employee_test with String fields: Employee ID, Employee Name, Email, Department, Location             | ServiceNow relational database tables                            |
| 6.          | Cloud Database         | Database service hosted with the ServiceNow cloud instance; stores import sets, employee records, transform history and reports                                 | ServiceNow Cloud (Personal Developer Instance)                   |
| 7.          | File Storage           | The Excel file is uploaded through Load Data and stored as an attachment; the sample sheet is prepared in Google Sheets and downloaded as .xlsx                 | ServiceNow attachment storage, Google Sheets / local file system |
| 8.          | External API-1         | No external API is called. Google Sheets is used only to prepare the sample data, which is exported as .xlsx                                                    | Google Sheets (file export)                                      |
| 9.          | External API-2         | Not applicable; the project uses a file-based import and needs no second external API                                                                           | N/A                                                              |
| 10.         | Machine Learning Model | Not applicable; the project is data import, transformation and reporting only                                                                                   | N/A                                                              |

|     |                                 |                                                                                                                                                                                       |                                                     |
|-----|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 11. | Infrastructure (Server / Cloud) | Application deployment on Cloud. Local: a computer with a web browser and the .xlsx file. Cloud server configuration: ServiceNow-hosted instance (no server setup needed by the team) | Cloud: ServiceNow Personal Developer Instance (PDI) |
|-----|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                                                                                                                                                                          | Technology                                                  |
|------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1.   | Open-Source Frameworks   | ServiceNow is a proprietary platform, so no open-source framework is used for the core logic. The solution relies on open formats and standards: Excel (.xlsx / Office Open XML), HTML, CSS and JavaScript           | Excel .xlsx format, HTML / CSS / JavaScript                 |
| 2.   | Security Implementations | Role-based access control with admin role, ACLs on tables and on pa_dashboards (Admin Overrides for dashboard creation), report sharing restricted by users, groups and roles, login authentication and HTTPS access | ServiceNow ACLs, Roles, Groups, HTTPS                       |
| 3.   | Scalable Architecture    | 3-tier architecture (browser presentation, ServiceNow application layer, database layer). A staging Import Set table and a reusable Transform Map allow larger spreadsheets to be loaded the same way                | ServiceNow Import Set, Transform Map, 3-tier cloud platform |
| 4.   | Availability             | Application runs on the ServiceNow cloud and is available through a browser anytime. Data stays safe in the staging table and the target table, and Transform History records every import                           | ServiceNow Cloud instance                                   |
| 5.   | Performance              | Staging table separates loading from transformation. Coalesce on Employee ID avoids duplicate records and unneeded inserts; reports use Count aggregation on a small table, so charts and lists open quickly         | Import Set staging, Coalesce, aggregate reports             |